

1. Administrative & Study Identification

Full Study Title: A Phase I/Ib Open-label, Multi-center Dose Escalation Study of JBH492 in Patients With Relapsed/Refractory Chronic Lymphocytic Leukemia (CLL) and Non-Hodgkin's Lymphoma (NHL) **Short Title/Acronym:** Safety and Preliminary Efficacy of JBH492 Monotherapy in Patients With CLL and NHL **Protocol Number:** CJBH492A12101 **NCT Number:** NCT04240704 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** JBH492 (Anti-CCR7 Antibody-Drug Conjugate) **Phase of Development:** Phase I/Ib **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, tolerability, pharmacokinetics (PK), immunogenicity, and preliminary efficacy of JBH492 as a single agent, and to determine the maximum tolerated dose (MTD) and recommended dose for future trials. **Secondary Objectives:** To evaluate the Overall Response Rate (ORR), Best Overall Response (BOR), Duration of Response (DOR), and Progression-Free Survival (PFS).

2.2 Background & Rationale To address the critical unmet medical need for patients with relapsed or refractory Chronic Lymphocytic Leukemia (CLL) and Non-Hodgkin's Lymphoma (NHL) who have exhausted standard-of-care therapies. The study investigates JBH492, an antibody-drug conjugate (ADC) targeting CCR7—a chemokine receptor frequently overexpressed in these hematological malignancies—with the aim of delivering targeted cytotoxicity directly to the tumor cells while sparing healthy tissues.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Dose escalation followed by expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 25 (Actual enrollment; study was terminated early due to strategic/business decisions, not safety concerns) **Number of Sites:** Multi-center **Estimated Study Start:** 07-Sep-2020 **Estimated Primary Completion:** 05-Sep-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of chronic lymphocytic leukemia (CLL) or histologically confirmed diagnosis of B- or T-cell non-Hodgkin's lymphoma (NHL). 2. Must have a site of disease amenable to biopsy, and be suitable and willing to undergo study required biopsies at screening and during therapy. 3. Must have received at least two (2) prior therapy regimens; patients with indolent lymphoma must have failed standard-of-care therapy or be intolerant/ineligible.

4.2 Exclusion Criteria 1. History of anaphylactic or other severe hypersensitivity/infusion reactions to ADCs or monoclonal antibodies, or prior treatment with maytansine (DM1 or DM4)-based ADCs. 2. Active or chronic corneal disorders, or active CNS involvement (unless effectively treated and local treatment was completed >4 weeks before first dose). 3. Known history of HIV, active HBV or HCV infection, or clinically significant cardiac disease/impaired cardiac function.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: JBH492 monotherapy administered via dose escalation (cohorts included 2.4 to 3.6 mg/kg every 3 weeks). **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Administered in 21-day cycles until disease progression, unacceptable toxicity, or discontinuation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures as medically indicated. **Prohibited:** Concurrent systemic anti-cancer therapies, other investigational agents, prophylactic intrathecal treatments (unless discontinued >5 half-lives prior to first dose).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Biopsy, Labs
Baseline	Day 0	First Dose (Cycle 1 Day 1), PK/PD baseline, Safety Assessments
Interim	Q3W (Every 3 Weeks)	Safety Labs, PK/ADA sampling, Efficacy Assessments (Tumor scans), Dosing
End of Study	EOT / Follow-up	Final Vital Signs, Exit Interview, Disease Progression tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs), Adverse Events (AEs), and Serious Adverse Events (SAEs) to establish the Maximum Tolerated Dose (MTD).

Measurement Method: Clinical safety observation, laboratory evaluations, and grading per CTCAE criteria.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR) assessed by investigators using iwCLL (CLL) or Lugano Classification (NHL). * Duration of Response (DOR). * Progression-Free Survival (PFS). * Best Overall Response (BOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous safety profiling; close monitoring for ADC-specific toxicities (e.g., corneal/ocular events, hematological toxicities like thrombocytopenia). **Serious Adverse Events (SAEs):** Standard 24-hour expedited reporting to the sponsor. **Data Monitoring Committee (DMC):** Internal dose-escalation review committee to evaluate DLTs prior to opening subsequent dose cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received ≥ 1 dose of JBH492) and Efficacy/Evaluable Set. **Sample Size Justification:** Target sample size was dynamic, guided by a Bayesian or standard Phase I dose-escalation model (final $N=25$ prior to early termination by the sponsor). **Handling of Missing Data:** Censoring applied for survival and PFS endpoints at the last adequate tumor assessment date.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of safety, PK data, and protocol compliance. **Audit Trail:** Robust electronic case report form (eCRF) system with data clarification pathways managed by Novartis.

11. Study Identifiers & Governance

Primary ID: CJBH492A12101 **Registry Number:** NCT04240704 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Phase I/Ib Open-label, Multi-center Dose Escalation Study of JBH492 in Patients With Relapsed/Refractory Chronic Lymphocytic Leukemia (CLL) and Non-Hodgkin's Lymphoma (NHL) **Official Regulatory Title:** Safety and Preliminary Efficacy of JBH492 Monotherapy in Patients With CLL and NHL

12. Clinical Framework (Disease Specific)

Primary Condition: Relapsed/Refractory Chronic Lymphocytic Leukemia (CLL) and Non-Hodgkin's Lymphoma (NHL) **Diagnosis Threshold:** Histologically confirmed B- or T-cell NHL or confirmed CLL with ≥ 2 prior therapies **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) Therapy **Optimization Target:** CCR7 receptor targeted cytotoxicity and determination of Maximum Tolerated Dose (MTD)

13. Study Procedures & Methodology

Management Pathway: Dose-escalation and expansion phase clinical trial pathway **Education Protocol (HCP):** Site initiation, safety protocol training regarding ADC-related toxicities (e.g., ocular/corneal exams) **Education Protocol (Patient):** Informed consent process, treatment compliance, and AE symptom self-monitoring **Quality Audit Mechanism:** Source data verification, eCRF monitoring, and Safety Review Committee evaluations

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 48 months (Actual primary completion Sept 2024) **Onsite Visit Cadence:** Every 3 weeks (Q3W) per dosing cycle **Remote Monitoring Frequency:** Continuous safety monitoring during the follow-up period **Checklist Review Interval:** End of each treatment cycle to assess DLTs prior to escalation

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						